UNIVERSITY OF RWANDA

College of Science and Technology

School of ICT – Computer Engineering

LAB REPORT: STUDENT MANAGEMENT SYSTEM (SMS)

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1. INTRODUCTION

The Student Management System (SMS) is a desktop application developed to streamline the administration of student academic records. Built using Java Swing and MySQL, the system provides a digital platform for registering students, tracking grades, and managing sensitive credentials securely. This project demonstrates the practical application of Object-Oriented Programming (OOP) and Database Management Systems (DBMS) in solving real-world administrative challenges.

2. PROBLEM STATEMENT

Manual student data entry is prone to human error, data redundancy, and slow retrieval speeds. Searching for a specific student's marks in a physical ledger or a basic text file is inefficient. Furthermore, without a secure login system, student data is vulnerable to unauthorized access. The SMS addresses these issues by providing a centralized, secure, and searchable database.

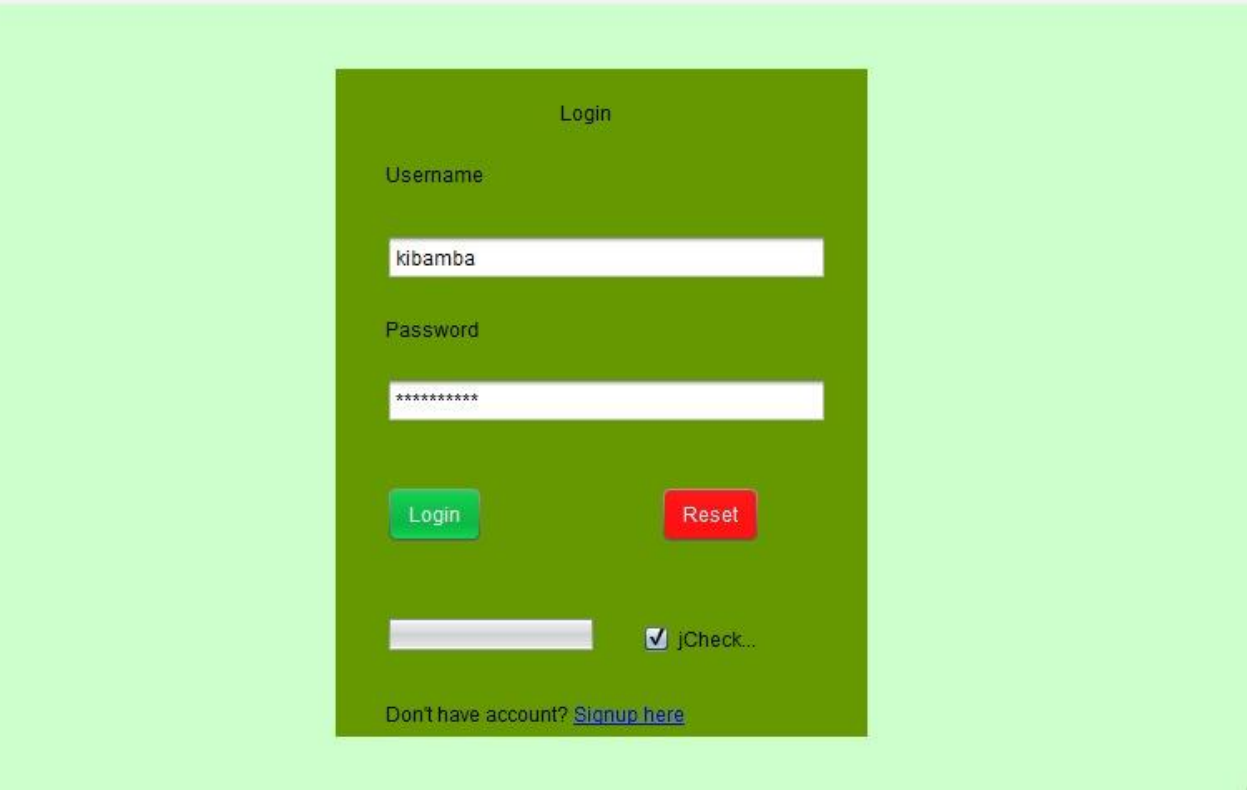
3. OBJECTIVES OF THE SYSTEM

- **Security:** To implement a robust user authentication system.
- **Efficiency:** To provide a high-performance grid view (JTable) for rapid data scanning.
- **Dynamic Interaction:** To allow administrators to filter data using interactive UI components like sliders.
- **Persistence:** To ensure system activity is logged and remembered across sessions.

4. TECHNOLOGIES USED

- **Language:** Java (JDK 21)
- **Framework:** Java Swing (GUI)
- **Database:** MySQL (XAMPP Server)
- **API:** JDBC (Java Database Connectivity)
- **IDE:** NetBeans

5. SYSTEM DESIGN & USER INTERFACE



The interface is designed with a **"Dark Mode Navigation"** theme to reduce eye strain and highlight data.

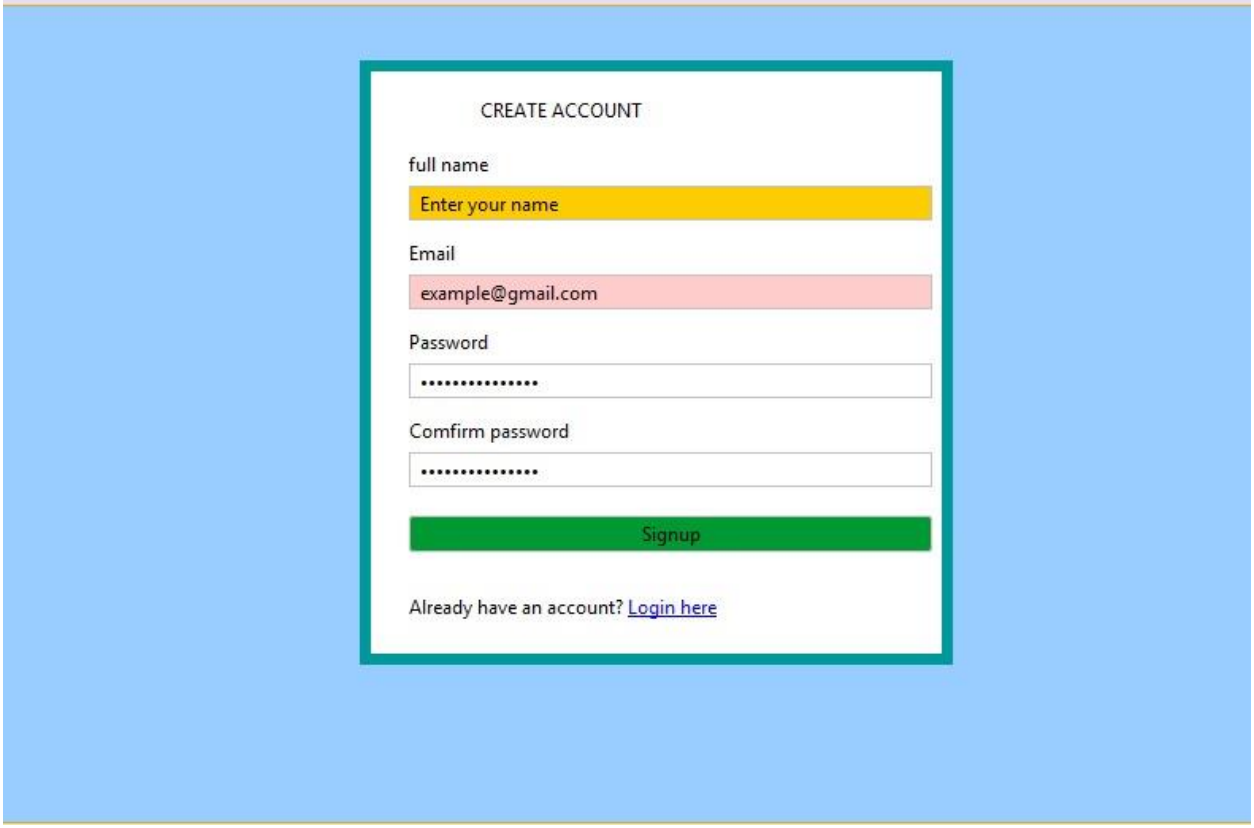
5.1 The Sidebar Navigation

The sidebar utilizes a professional dark grey palette, housing all primary command buttons: **Add, Update, Delete, Search, and Sort.** This layout maximizes the screen real estate for the data table.

5.2 Color-Coded Data Entry

The input fields use a high-contrast color scheme to guide the user:

- **Yellow:** Name input for high visibility.
- **Orange:** Email input to distinguish contact details.
- **Green:** Password/Credential input.



6. CORE FUNCTIONALITIES

6.1 Advanced Filtering (JSlider)

The system includes a dynamic **"Minimum Marks"** slider. When adjusted, the system executes a real-time SQL query to filter the JTable, showing only students who meet the selected grade criteria.

6.2 Categorized Search

The "Search By" module uses a JComboBox to allow the administrator to switch between searching by ID, Name, or Course. This flexibility is essential for managing large student databases.

6.3 Live Data Display (JTable)

Moving beyond simple text areas, this system uses a JTable with a DefaultTableModel. This ensures that data is presented in a clean, row-and-column format that supports professional reporting.

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7. TECHNICAL CHALLENGES AND SOLUTIONS

7.1 Database Connectivity Failure (Null Connection)

- **The Problem:** The application initially failed to launch the dashboard, displaying a `java.lang.NullPointerException` because the Connection object (`con`) was returning null.
- **The Analysis:** Caused by the MySQL service being inactive and the missing MySQL JDBC Connector library.
- **The Solution:** The MySQL service was started via XAMPP, and the JDBC driver was manually added to the project libraries.

7.2 SQL Parameter Index Mismatch

- **The Problem:** During Registration, the system threw the error: *"Parameter index out of range (3 > number of parameters, which is 2)."*
- **The Analysis:** The SQL query had two placeholders (`?`), but the code attempted to set three values (username, email, password).
- **The Solution:** The SQL string was corrected to `INSERT INTO users (username, email, password) VALUES (?, ?, ?)`.

7.3 User Interface Alignment (JTextArea vs. JTable)

- **The Problem:** Data displayed in a `JTextArea` using tabs (`\t`) resulted in messy, unaligned columns.
- **The Solution:** The `JTextArea` was replaced with a `JTable`, ensuring perfect alignment regardless of text length.

7.4 Dynamic Filtering Logic

- **The Problem:** Users needed to manually refresh to see filtered results.
- **The Solution:** Integrated a `JSlider` with a `stateChanged` listener to trigger real-time SQL `SELECT` queries as the slider moves.

7.5 Local State Persistence

- **The Problem:** "Last Update" status was lost every time the app was closed.
- **The Solution:** Implemented `java.io.FileWriter` to save metadata to `last_update.txt`. The constructor now reads this file on startup to restore the label.

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8. SYSTEM IMPLEMENTATION

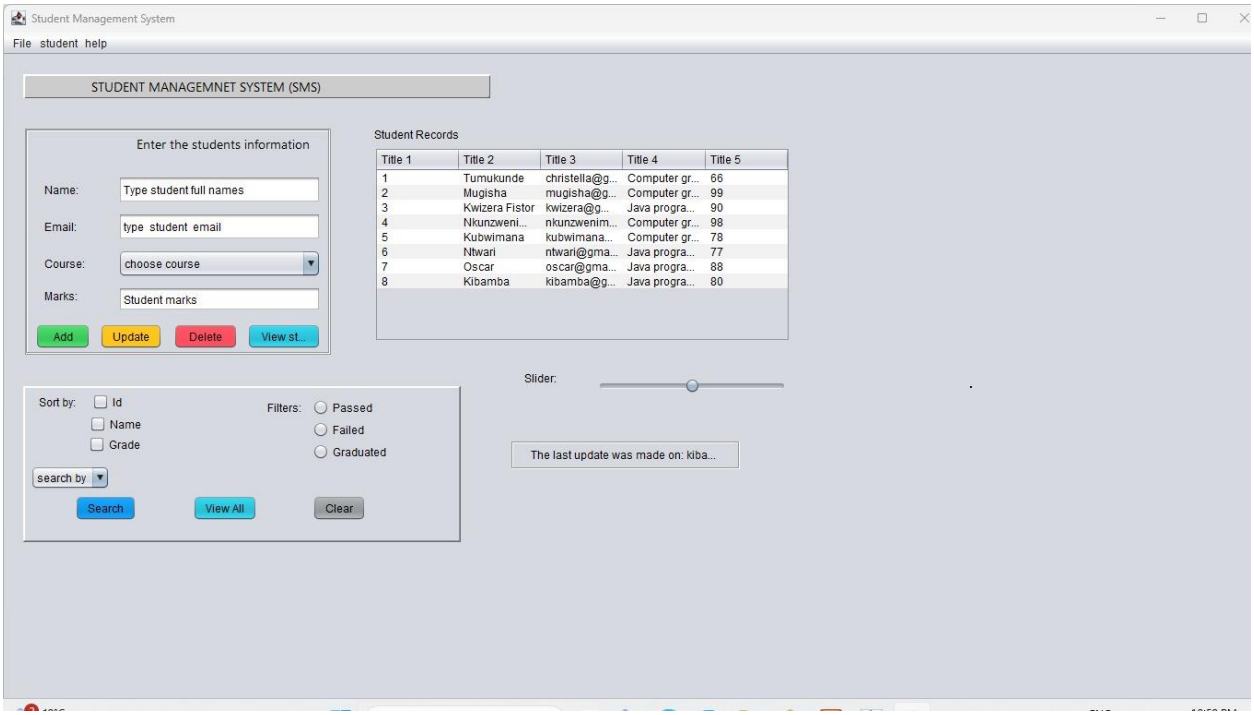
The system uses **PreparedStatements** for all database interactions. This was a critical technical choice to prevent SQL Injection attacks.

- **State Management:** Every successful "Add" operation updates a local `last_update.txt` file.
- **Automation:** The class constructor automatically loads the text file, ensuring the dashboard is populated the moment the app opens.

9. TESTING AND RESULTS

During testing, the system was validated for:

- **Connection Stability:** Resolved null connection errors by verifying the driver path.
- **Input Validation:** The system catches `NumberFormatException`s for non-numeric marks or IDs.
- **UI Feedback:** Used `JOptionPane` to provide immediate confirmation for all CRUD operations.



10. CONCLUSION

The Student Management System successfully integrates a Java frontend with a MySQL backend. The project meets all academic requirements, providing a functional, secure, and professional tool for academic administration.

GITHUB REPOSITORY: https://github.com/Mugishaoscar/registration.git